

42.510 Statistical Learning for Data Science

Multiple Testing

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Week 12, Lecture 1

Outline

- Overview
- Recall: Hypothesis testing
- Multiple testing: challenges
- Family-wise Error Rate (FWER)
 - ▶ Bonferroni correction
 - ▶ Holm's method

Next class

- False Discovery Rate (FDR)
 - ▶ Benjamini-Hochberg procedure
- Re-sampling: p -values and FDR

Overview

- Statistical learning mostly addresses *estimation* and *prediction*.
- Here we cover another aspect: **testing**
 - ▶ Traditionally we *test* single hypothesis, i.e., a regression coefficient is zero, or, two means are equal, etc.
 - ★ H_0 : the expected blood pressures of mice in the control and the treatment groups are the same.
 - ▶ With huge amounts of data, we sometimes wish to test multiple hypotheses together:

$$H_{01}, H_{02}, \dots, H_{0m}.$$

- ▶ H_{0j} : the expected values of the j -th biomarker among mice in the control and the treatment groups are equal.

Hypothesis testing: a review

- Hypothesis tests allow us to answer simple “yes-or-no” questions, such as:
 - ▶ Is the true coefficient β_j in a linear regression equal to zero?
 - ▶ Does the expected blood pressure among mice in the treatment group equal the expected blood pressure among mice in the control group?
- Hypothesis testing proceeds as follows:
 - 1 Define the null and alternative hypotheses.
 - 2 Construct the test statistic.
 - 3 Compute the p -value.
 - 4 Decide whether to reject the null hypothesis.

Hypothesis testing: a review

Null and Alternate hypotheses

- We divide the world into **null** and **alternative** hypotheses.
- **Null hypothesis**, denoted H_0 : in general, *status quo* or default state, hypothesis we are seeking evidence against.
 - ▶ The true coefficient β_j equals zero.
 - ▶ There is no difference in the expected blood pressures.
- **Alternative hypothesis**, denoted H_1 or H_A : new belief, hypothesis which requires evidence for.
 - ▶ The true coefficient β_j is non-zero.
 - ▶ There is a difference in the expected blood pressures.

Hypothesis testing: a review

Test statistic

A *test statistic* is a function of the sample data on which the decision (reject $\mathbf{H}_0$ or do not reject $\mathbf{H}_0$) is to be based.

- In the *example discussed*:

- ▶ Let $x_1^t, \dots, x_{n_t}^t$ denote the blood pressure measurements for the n_t mice in the treatment group, and let $x_1^c, \dots, x_{n_c}^c$ denote the blood pressure measurements for the n_c mice in the control group.
- ▶ Let $\mu_t = \mathbb{E}(X_1^t)$ and $\mu_c = \mathbb{E}(X_1^c)$ and denote:

$$\hat{\mu}_t = \bar{x}_t = \frac{1}{n_t} \sum_{i=1}^{n_t} x_i^t, \quad \hat{\mu}_c = \bar{x}_c = \frac{1}{n_c} \sum_{i=1}^{n_c} x_i^c,$$

$$s = \sqrt{\frac{(n_t - 1)s_t^2 + (n_c - 1)s_c^2}{n_t + n_c - 2}},$$

$$s_t^2 = \frac{1}{n_t - 1} \sum_{i=1}^{n_t} (x_i^t - \bar{x}_t)^2, \quad s_c^2 = \frac{1}{n_c - 1} \sum_{i=1}^{n_c} (x_i^c - \bar{x}_c)^2.$$

Hypothesis testing: a review

Test statistic

- To test the hypothesis $\mathbf{H}_0 : \mu_t = \mu_c$, we use a two-sample t -statistic with equal variances (see slides/video uploaded earlier):

$$T = \frac{\hat{\mu}_t - \hat{\mu}_c}{s \sqrt{\frac{1}{n_t} + \frac{1}{n_c}}}.$$

- It is known that if $\mathbf{H}_0$ holds, then $T \sim t$ -distribution with $n_t + n_c - 2$ degrees of freedom (d.f.).
- For $n > 40$ the t -distribution with n d.f. is well-approximated by $\mathcal{N}(0, 1)$.

Hypothesis testing: a review

p -value

- The p -**value** is the probability of observing a test statistic at least as extreme as the observed statistic or worse, under the assumption that $\mathbf{H}_0$ is true.
- A small p -value provides evidence against $\mathbf{H}_0$.
- Suppose we obtain $T = 2.5$ for the above test $\mathbf{H}_0 : \mu_T = \mu_C$.
- Assuming $n_t + n_c - 2 > 40$, under $\mathbf{H}_0$, we have $T \sim \mathcal{N}(0, 1)$.
- Hence p -value is $\mathbb{P}(|T| \geq 2.5) = 0.012$ which is quite small.

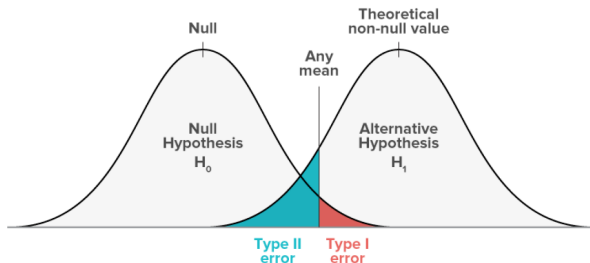
p-value

<u>P-VALUE</u>	<u>INTERPRETATION</u>
0.001	HIGHLY SIGNIFICANT
0.01	
0.02	
0.03	
0.04	SIGNIFICANT
0.049	
0.050	OH CRAP. REDO CALCULATIONS.
0.051	ON THE EDGE OF SIGNIFICANCE
0.06	
0.07	HIGHLY SUGGESTIVE, SIGNIFICANT AT THE $P < 0.10$ LEVEL
0.08	
0.09	
0.099	HEY, LOOK AT THIS INTERESTING SUBGROUP ANALYSIS
≥ 0.1	

Source: XKCD

Hypothesis testing: Type I and Type II error

- **Type I error:** Rejecting H_0 when H_0 is true.
- **Type II error:** Not rejecting H_0 when H_A is true.



- Naturally we would want both errors to be small, which may not be possible to do together. Achieving both simultaneously is hard unless sample size is increased.

Hypothesis testing: decision

- We call the **level of significance** of a test α if $\mathbb{P}(\text{Type I error}) \leq \alpha$.
 - We reject $\mathbf{H}_0$ if p -value falls below this level of significance α : often chosen as 0.05, 0.01, 0.001.
 - This is when we believe we have made a *discovery*.
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- What happens if we have many many such hypotheses?

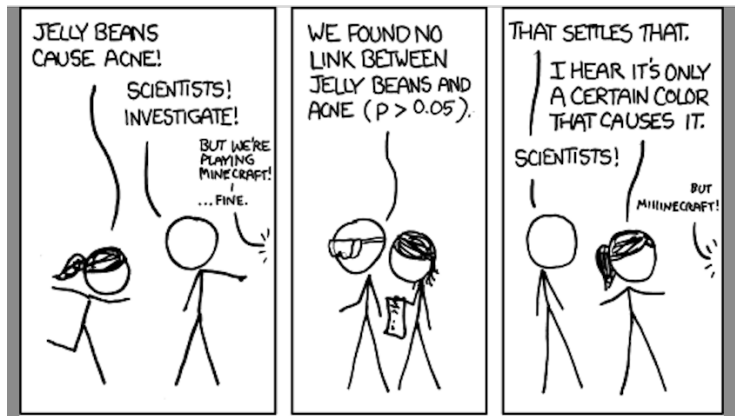
Multiple testing

- Suppose we want to test m null hypotheses (simultaneously):

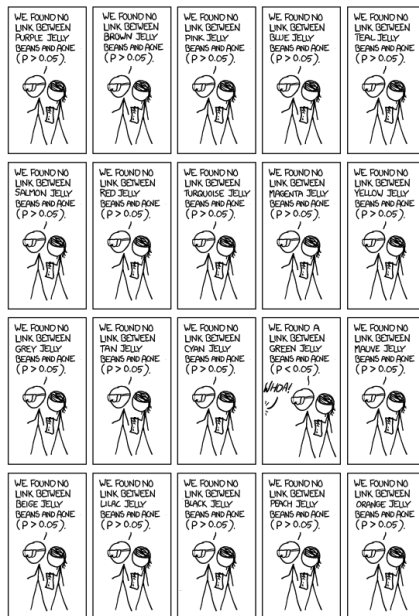
$$\mathbf{H}_{01}, \mathbf{H}_{02}, \dots, \mathbf{H}_{0m}.$$

- Can we reject all hypotheses for which p -value falls below say 0.05?
- What is our Type I error here?

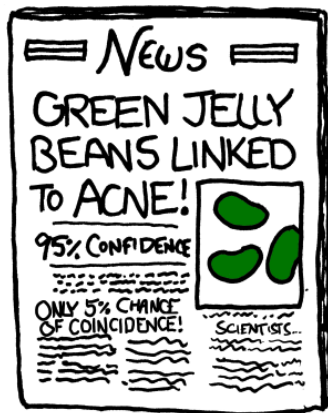
Multiple testing



Multiple testing



Multiple testing



Source: XKCD

Multiple testing

Example: Toss a fair coin ten times and test for fairness at level 0.005.

- $\mathbb{P}(\text{all heads or all tails}) = 0.00195$.
- Hence we may reject the null hypothesis of *fairness* if this happens.
- Now conduct the same experiment with 1024 coins.
- What would you observe?
- Perhaps one or two coins show all heads or all tails.
- Would you reject that the null hypothesis for those cases?

The Challenge of Multiple testing

- Suppose we test

$$H_{01}, H_{02}, \dots, H_{0m}.$$

- Reject any null hypothesis with a p -value below $\alpha = 0.01$.
- Suppose all the hypotheses are actually true.
- Then we expect to falsely reject approximately $0.01 \times m$ null hypotheses.
- If $m = 10000$, then we expect to falsely reject 100 null hypotheses by chance.
- That is a lot of false positives!!

Family-Wise Error Rate (FWER)

- The **Family-Wise Error Rate** (FWER) is the probability of making at least one Type I error when conducting m hypothesis tests.
- Then $\text{FWER} = \mathbb{P}(V \geq 1)$ in the following table of counts from the m hypotheses tests.

	H_0 is True	H_0 is False	Total
Reject H_0	V	S	R
Do not Reject H_0	U	W	$m - R$
Total	m_0	$m - m_0$	m

How to control FWER?

$$\begin{aligned}\text{FWER} &= 1 - \mathbb{P}(\text{do not falsely reject any null hypotheses}) \\ &= 1 - \mathbb{P}\left(\bigcap_{j=1}^m \{\text{do not false reject hypothesis } \mathbf{H}_{0j}\}\right).\end{aligned}$$

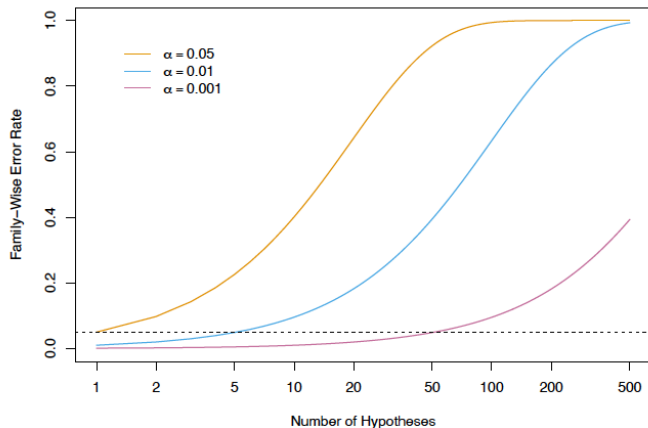
- If the tests are independent and all $\mathbf{H}_{0j}$ are true then

$$\text{FWER} = 1 - \prod_{i=1}^m (1 - \alpha) = 1 - (1 - \alpha)^m.$$

- Let us plot FWER for different levels of $\alpha = 0.05, 0.01, 0.001$ for increasing m .

How to control FWER?

$$\text{FWER} = 1 - \prod_{i=1}^m (1 - \alpha) = 1 - (1 - \alpha)^m.$$



Bonferroni correction for controlling FWER

- Let A_j denote the event that we falsely reject the j^{th} null hypothesis. Then

FWER = $\mathbb{P}$ (falsely rejecting at least one null hypotheses)

$$= \mathbb{P}\left(\bigcup_{j=1}^m A_j\right) \leq \sum_{j=1}^m \mathbb{P}(A_j)$$

- If we only reject a hypothesis when its p -value is less than α/m then

$$\text{FWER} \leq \sum_{j=1}^m \mathbb{P}(A_j) \leq \sum_{j=1}^m \frac{\alpha}{m} = m \times \frac{\alpha}{m} = \alpha.$$

since, under null hypotheses $\mathbb{P}(A_j) \leq \alpha/m$.

- We call this the **Bonferroni correction**:
 - For controlling FWER at level α , we reject any null hypothesis with p -value below α/m .

Example (controlling FWER): Fund manager data

Manager	Mean, $\bar{x}$	s	t -statistic	p -value
One	3.0	7.4	2.86	0.006
Two	-0.1	6.9	-0.10	0.918
Three	2.8	7.5	2.62	0.012
Four	0.5	6.7	0.53	0.601
Five	0.3	6.8	0.31	0.756

- H_{0j} : the j^{th} manager's expected excess return equals zero.
- If we reject H_{0j} if the p -value is less than $\alpha = 0.05$, then we will conclude that the first and third managers have significantly non-zero excess returns.
- However, we have tested multiple hypotheses, so the FWER is greater than 0.05.

Example (controlling FWER): Fund manager data

Controlling with Bonferroni correction

Manager	Mean, $\bar{x}$	s	t -statistic	p -value
One	3.0	7.4	2.86	0.006
Two	-0.1	6.9	-0.10	0.918
Three	2.8	7.5	2.62	0.012
Four	0.5	6.7	0.53	0.601
Five	0.3	6.8	0.31	0.756

- Using a Bonferroni correction, we reject for p -values less than $\alpha/m = 0.05/5 = 0.01$.
- Consequently, we will reject the null hypothesis only for the first manager.
- Now the FWER is at most 0.05.

Holm's (step-down) method for controlling FWER

- Specify α , the level at which to control FWER.
- Compute p -values $p_1, \dots, p_m$ for the m null hypotheses $\mathbf{H}_{01}, \mathbf{H}_{02}, \dots, \mathbf{H}_{0m}$.
- Order the m p -values so that $p_{(1)} \leq \dots \leq p_{(m)}$.

- Define

$$L = \min \left\{ j : p_{(j)} > \frac{\alpha}{m+1-j} \right\}.$$

- Reject all null hypotheses $\mathbf{H}_{0j}$ for which $p_j < p_{(L)}$.
 - ▶ This is called **Holm's method** for controlling FWER at level α .

Example (controlling FWER): Fund manager data

Controlling with Holm's method

Manager	Mean, $\bar{x}$	s	t -statistic	p -value	order of p -value
One	3.0	7.4	2.86	0.006	$p_{(1)}$
Two	-0.1	6.9	-0.10	0.918	$p_{(5)}$
Three	2.8	7.5	2.62	0.012	$p_{(2)}$
Four	0.5	6.7	0.53	0.601	$p_{(3)}$
Five	0.3	6.8	0.31	0.756	$p_{(4)}$

- The ordered p -values are

$$p_{(1)} = 0.006, p_{(2)} = 0.012, p_{(3)} = 0.601, p_{(4)} = 0.756, p_{(5)} = 0.918.$$

- Hence using Holm's method we reject the first two null hypotheses (with respect to managers One and Three):

$$p_{(1)} = 0.006 < 0.05/(5 + 1 - 1) = 0.0100,$$

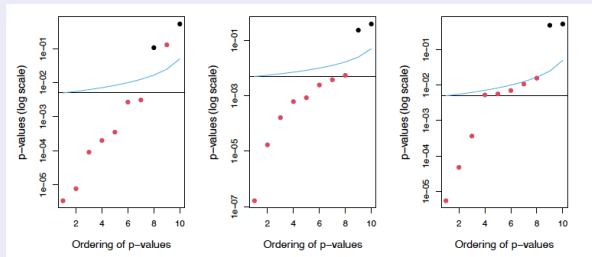
$$p_{(2)} = 0.012 < 0.05/(5 + 1 - 2) = 0.0125,$$

$$p_{(3)} = 0.601 > 0.05/(5 + 1 - 3) = 0.0167.$$

Bonferroni vs. Holm's

Comparing $m = 10$ p -values for $m = 10$ null hypotheses

- 2 true H_0 (black) and 8 false H_0 (red).



- Each panel displays a separate simulation with sorted p -values.
- We want to control FWER at level 0.05.
- Bonferroni rejects H_0 with p -values below the black line.
- Holm's rejects H_0 with p -values below the blue line.
- Clearly Holm's often rejects more than Bonferroni, but to a varying degree.

Bonferroni vs Holm's

- Bonferroni is simple . . . reject any null hypothesis with a p -value below α/m .
- Holm's method is a bit more complicated, but it will lead to more rejections while controlling FWER!!
- So, Holm's is clearly a **better** choice!

Other methods

- There are lots of specialized approaches to control FWER.
- For example:
 - ▶ *Tukey's Method*: for pairwise comparisons of the difference in expected means among a number of groups.
 - ▶ *Scheffe's Method*: for testing arbitrary linear combinations of a set of expected means, e.g.

$$\mathbf{H}_0 : \frac{1}{2}(\mu_1 + \mu_2) = \frac{1}{3}(\mu_2 + \mu_4 + \mu_5)$$

- Bonferroni and Holm are general procedures that will work in most settings. However, in certain special cases, methods such as Tukey and Scheffe can give better results: i.e., more rejections while maintaining FWER control.

References

- James et al. (2023) *An Introduction to Statistical Learning: with Applications in R, Second edition*, Chapter 13.